

ASSIGNMENT 4

MMATH FIRST YEAR, 2025

Problems

1. Show that if g_1 and g_2 are primitive roots for p , a prime, then g_1g_2 is not.
2. If $m > 6$ and m has primitive roots, then show that the product of all the primitive roots is congruent to 1 modulo m .
3. Find all n such that $2^n \mid (3^n - 1)$. Explain your solution.
4. Let g be a primitive root for p^n for some $n > 1$, where p is a prime. Show that g is also a primitive root for p .
5. Suppose g is a primitive root for $n > 2$. What is the smallest positive integer k such that $g^k \equiv -1 \pmod{n}$?
6. Show that for a prime p , the sum of the distinct primitive roots for p is congruent to $\mu(p-1)$ modulo p .
7. Suppose f is multiplicative. Show that f is completely multiplicative iff $f^{-1}(n) = \mu(n)f(n)$ for every n .
8. Find all positive integers n such that (i) $\phi(n) = n/2$, (ii) $\phi(n) = 2n$, (iii) $\phi(n) = 12$.
9. Show that a quadratic residue for a prime p cannot be a primitive root for p .
10. Given $q > 1$, how many characters are there modulo q and how many of them are real valued? Justify your answer.
11. Suppose $q_1, q_2 > 1$ and χ_1 and χ_2 are two characters of moduli q_1 and q_2 .
 - (a) Show that $\chi_1\chi_2$ is a character itself of modulus $[q_1, q_2]$.
 - (b) If q_1 and q_2 are coprime and χ is a character of modulus $q = q_1q_2$, then show that χ can be expressed uniquely as a product of two characters $\chi_1 \pmod{q_1}$ and $\chi_2 \pmod{q_2}$.
12. Describe all the Dirichlet characters modulo 2, 3, 4, 5, 6, 9, 12, 30.

13. Show that

$$\sum_{k|n} d(k)^3 = \left(\sum_{k|n} d(k) \right)^2.$$

14. Let h denote the indicator function for squares of integers. Express h using the Möbius function μ .

15. Show that

$$\sum_{\substack{1 \leq a \leq n \\ (a,n)=1}} e^{2\pi i \frac{a}{n}} = \mu(n).$$

16. (i) For which integers n , is $d(n)$ odd?
(ii) For which n , $\sigma(n)$ is odd?

17. Show that $\phi(n) > n/6$ if n has at most 8 distinct prime factors.

18. Show that

$$\frac{n}{\phi(n)} = \sum_{d|n} \frac{\mu^2(d)}{\phi(d)}.$$

19. Show that the sum

$$\sum_{d^2|n} \mu(d)$$

is 1 or 0 according as n is square-free or not.

20. Let $\lambda : \mathbb{N} \rightarrow \mathbb{C}$ be a function that satisfies

$$\lambda(m)\lambda(n) = \sum_{d|(m,n)} \lambda(mn/d^2) \text{ for all integers } m, n \geq 1.$$

Show that for all positive integers m, n ,

$$\lambda(mn) = \sum_{d|(m,n)} \mu(d)\lambda(m/d)\lambda(n/d).$$

21. Show that for any complex numbers $c_1, c_2, \dots, c_q$,

$$\sum_{\chi \pmod{q}} \left| \sum_{1 \leq n \leq q} c_n \chi(n) \right|^2 = \phi(q) \sum_{\substack{1 \leq n \leq q \\ (n,q)=1}} |c_n|^2.$$

Hint: Expand the square, bring the sum over characters inside and use orthogonality.

22. Let $\chi \pmod{p}$ be a non-principal Dirichlet character modulo p , a prime. Show that

$$\left| \sum_{a \pmod{p}} \chi(a) e^{2\pi i \frac{a}{p}} \right| = \sqrt{p}.$$

Hint: Consider the diagonal and the off-diagonal separately after expanding the square.

23. Show that the sum

$$S_\chi = \sum_{1 \leq n \leq q} n \chi(n)$$

is an integer for any character $\chi \pmod{q}$.